...mputer and Emerging Sciences
Lahore Campus

Programming Fundamental (CL1002)

Sessional-II Exam

Date: Oct 14th 2025

Course Instructor(s)

Mr. Aqib Zeeshan

Total Time (Hrs): 2:20
Total Marks: 40
Total Questions: 3

[redacted] Roll No   [redacted] Section   [redacted] Student Signature

Do not write below this line

Attempt all the questions.

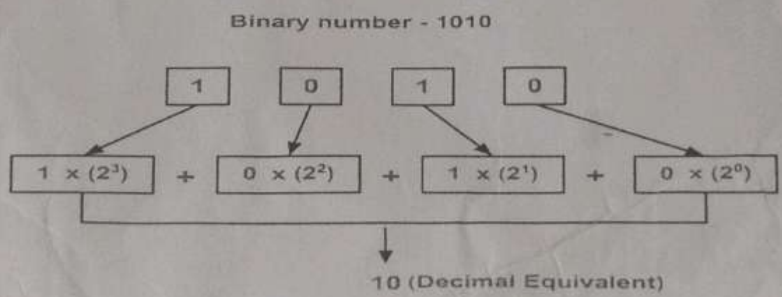
This is an individual Exam and you are supposed to complete it in 02 hours 30 Minutes. Please adhere to the following guidelines:-

- **You are in exam conditions, not communication is allowed. If will cancel your paper and mark your absent in case of any problems**
- **Understanding is part of this exam, do not ask any questions, if you have confusions, assume things, and write your assumptions, No quires, means no quires. Asking for teacher is not allowed.**
- **No extra time will be awarded.**
- **Submission of exam in the correct folder is your responsibility, if you submit in the wrong folder or submit the wrong file, only you will be responsible and no excuses will be accommodated.**
- **Please Submit your exam in following folder//exam/ *Fall 2025 Mid Exam\"Subject Name"\"Section Name" \submissions***
- **No help from internet is allowed.**
- **Cheating cases will be reported to DC.**

CLO #: CLO statement for question Q1 ....

Q1: Write a program that takes a binary number and convert to decimal number. [marks: 15]

Example 1:



Example 2: ***Input***: *n = 12*
Output: *"1100"*

Explanation: *the binary representation of 12 is "1100", since* $12 = 1\times2^3 + 1\times2^2 + 0\times2^1 + 0\times2^0 = 12$

CLO #: CLO statement for question Q2 ....

Q2: Write a program that finds the sum of series. [marks: 15]

$$1+\frac{x}{1!}+\frac{x^2}{2!}+\frac{x^3}{3!}+\cdots+\frac{x^n}{n!}$$

?

Example 1:
x=1
n=5
1 + 1/2 + 1/6 + 1/24 + 1/120 = 1.7167
Example 2:
X=1
N= 3
Output: 1.6667
Explanation: 1 + 1/2 + 1/6 = 1.6667

CLO #: CLO statement for question Q3 ....

Q3: Palindromic Prime Pyramid [marks : 10]
Write a program that prints the first n palindromic prime numbers in a centered pyramid pattern.

A palindromic prime is a number that is:

Prime Number: The number which is only divisible by 1 and it self.

A palindrome (reads the same backward)

Example output (for n = 6):

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   2
  3 5
 7 11 101
```

For n = 6, you'll need: 2, 3, 5, 7, 11, 101